



Subject Code	CSC 1017
Subject Name	SYSTEM ANALYSIS AND DESIGN
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Assessment number &title	Part two - Designing an AI-Enabled Community Health Services Management System (AI-OCHSMS)
I declare that all material in this assignment/project/tutorial is my own work except where there is clear acknowledgement or reference to the work of others and I have complied and agreed to the University statement on Plagiarism and Academic Integrity. Signature: _____ Date: 12/10/2025 	

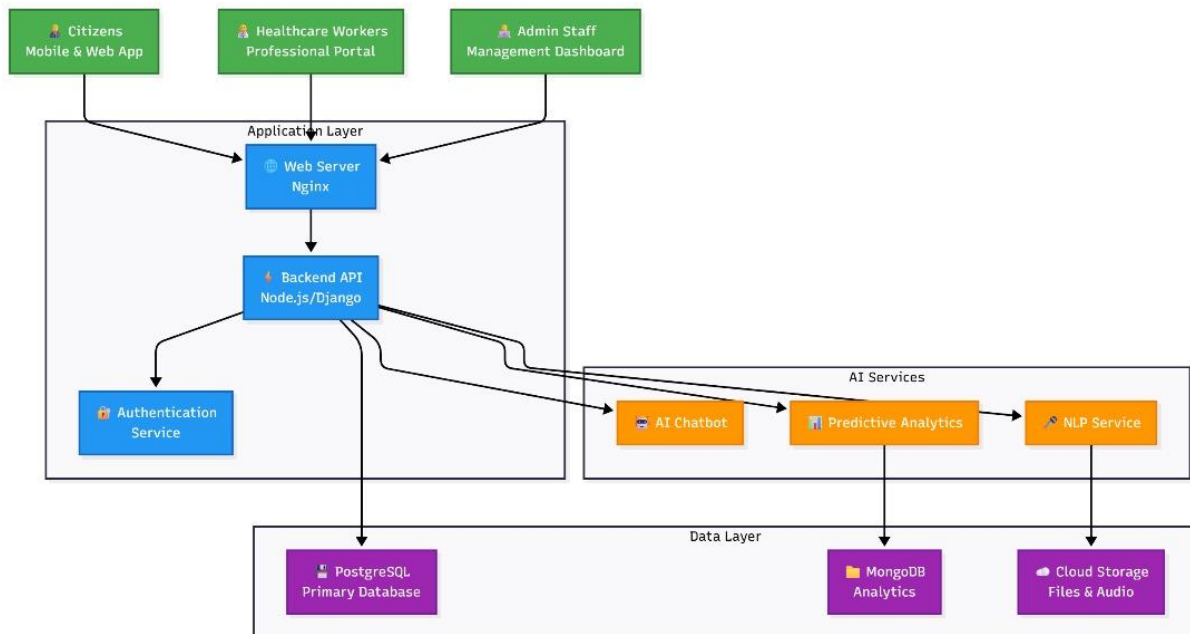
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1. System Architecture Diagram

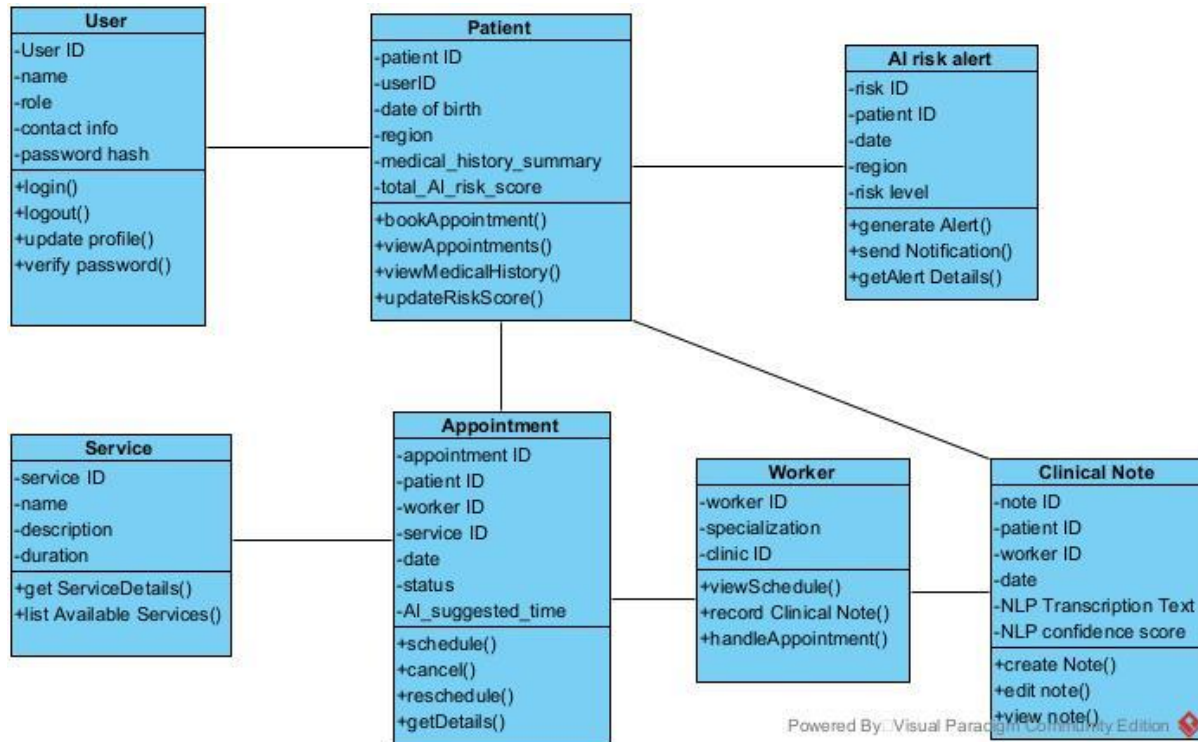
We chose a **Multi-Layer Service-Oriented Architecture** to clearly separate the user interface, core application logic, and specialized AI services. This modular design, using components like **Nginx**, **Node.js/Django**, and separate databases (**PostgreSQL** and **MongoDB**), ensures the system is strong, secure, and highly scalable.



Layer Name	Role (What it Does)	Key Components (Based on Your Image)
Presentation Layer	The interfaces that users interact with.	Citizens Mobile & Web App, Healthcare Workers Professional Portal, Admin Staff Management Dashboard.
Application Layer	Manages core business logic, user requests, and directs data flow.	Web Server (Nginx), Backend API (Node.js/Django), Authentication Service.
AI Services Layer	Runs all the specialized smart models. It is separated for flexibility and processing power.	AI Chatbot, Predictive Analytics, NLP Service.
Data Layer	Stores all the system's data securely.	PostgreSQL Primary Database (Structured Data), MongoDB Analytics (Unstructured data/Logs), Cloud Storage (Files/Audio).

2. Detailed Class Diagrams

This diagram shows the main objects (classes) in our system. It is a direct blueprint of our database (ERD) and defines the behavior of the system.



The diagram is organized around the data entities and includes specialized methods to manage the AI features:

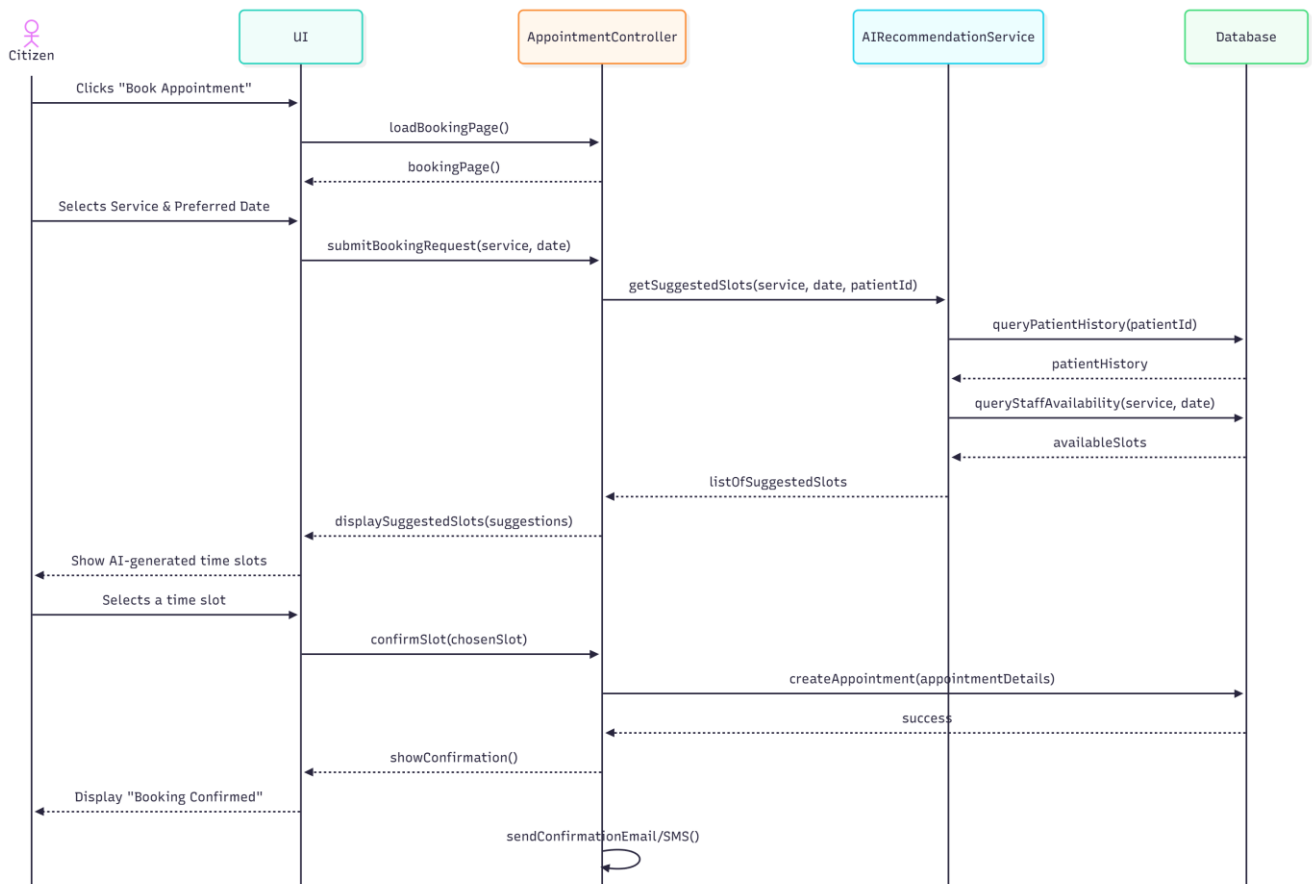
- **Patient Class:** Includes the method **updateRiskScore()** and the attribute **total AI risk score**, proving the class is ready for the predictive output.
- **Clinical Note Class:** Includes the attributes of **NLP Transcription Text** and **NLP confidence score**, which store the data generated by the voice-to-text AI.
- **Appointment Class:** Includes the attribute **AI_suggested_time**, demonstrating the integration of the AI recommendation feature directly into the booking data.

3. Sequence Diagrams (3 Use Cases)

These diagrams show the exact steps of data flow in our most complex, AI-driven processes.

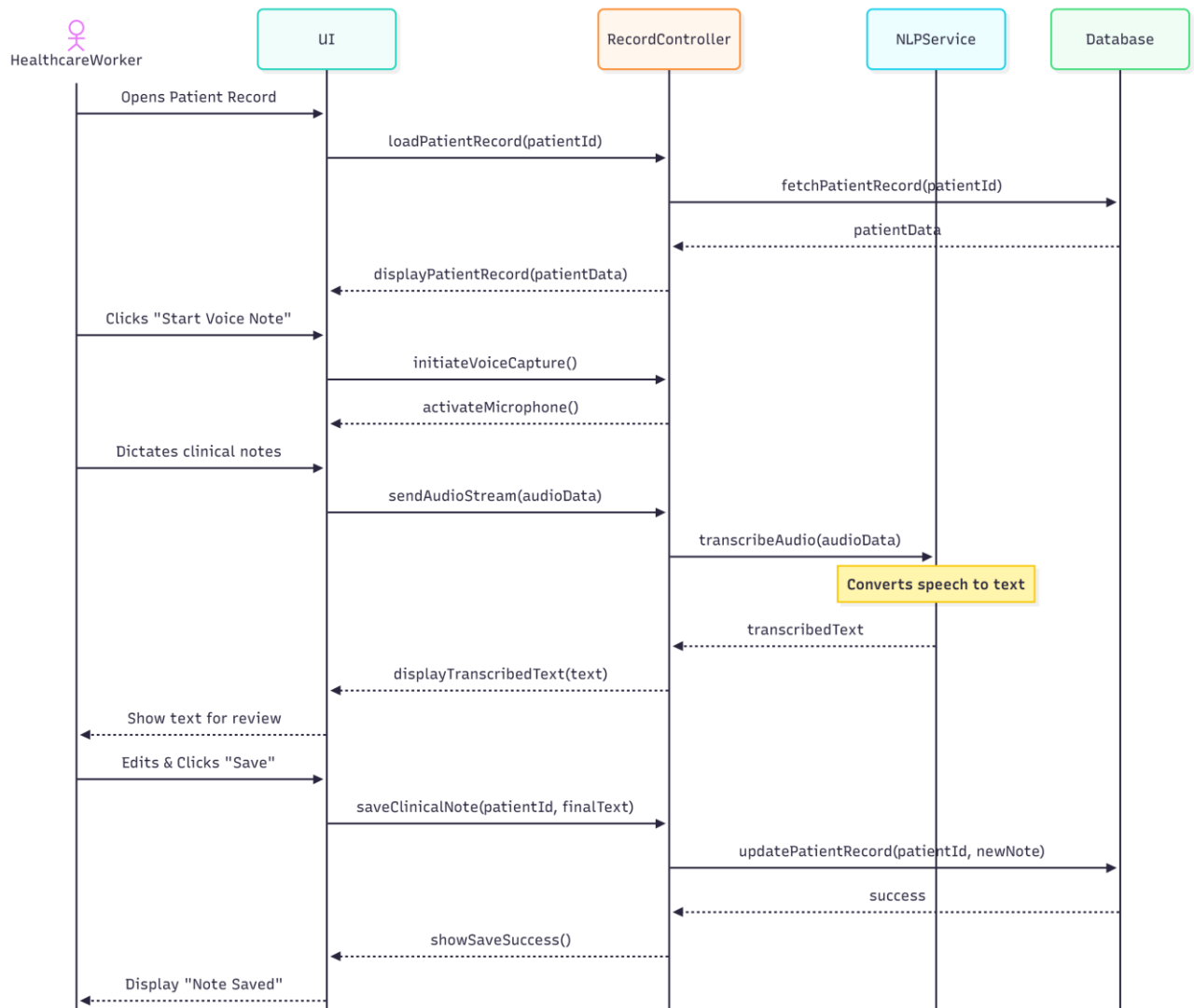
Sequence 3.1: Citizen Books Appointment (with AI)

This proves that the system uses the AI to choose the best time slot.



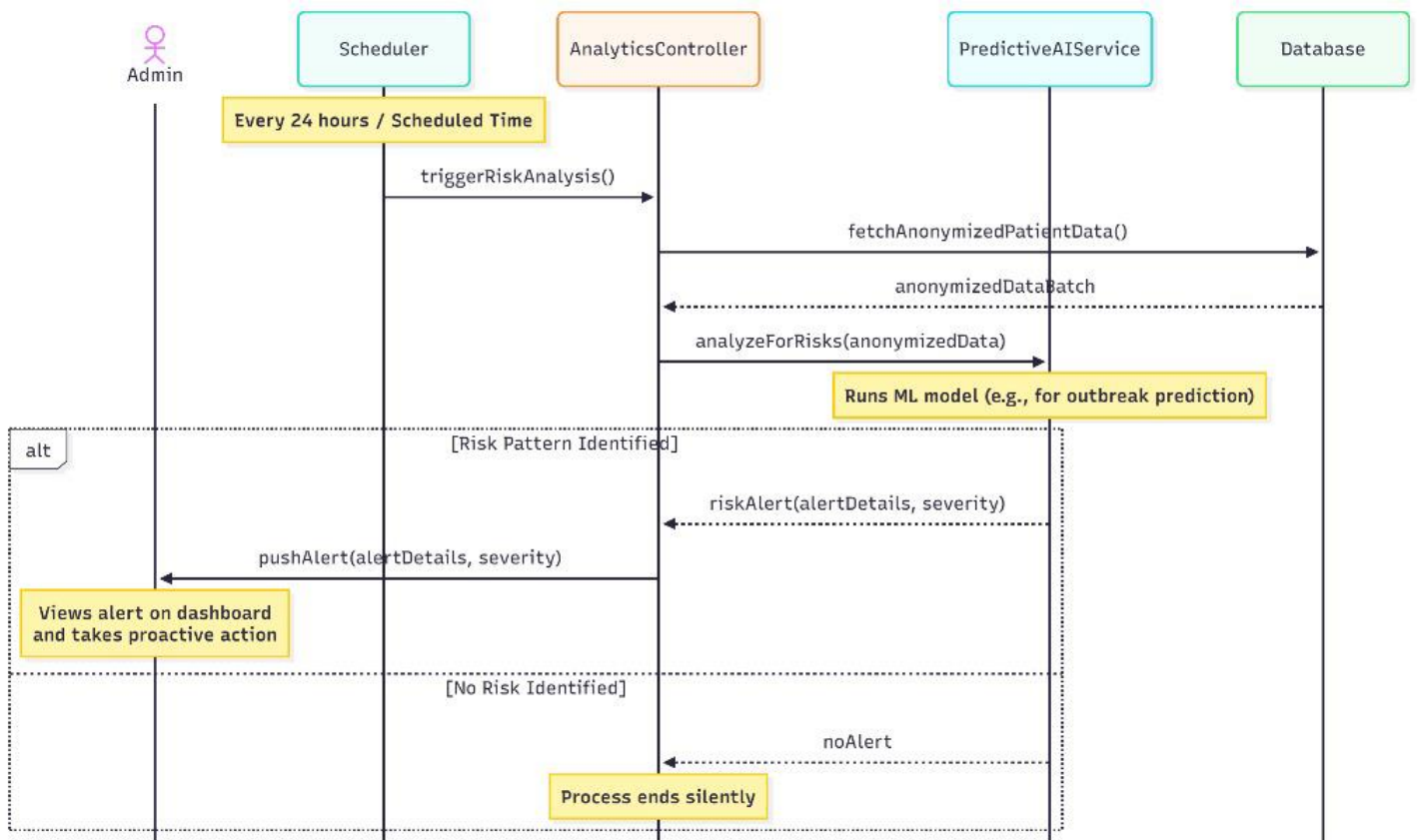
Sequence 3.2: Healthcare Worker Saves Voice Note (with NLP)

This proves the system converts speech to text for efficiency.



Sequence 3.3: System Generates Predictive Risk Alert

This proves the system automatically checks for outbreaks and notifies the Admin Staff



4. ERD / Relational Schema

The database design is the foundation of our system. It uses **Information Engineering (Crow's Foot) Notation** and is designed to store both patient data and AI-generated results securely.

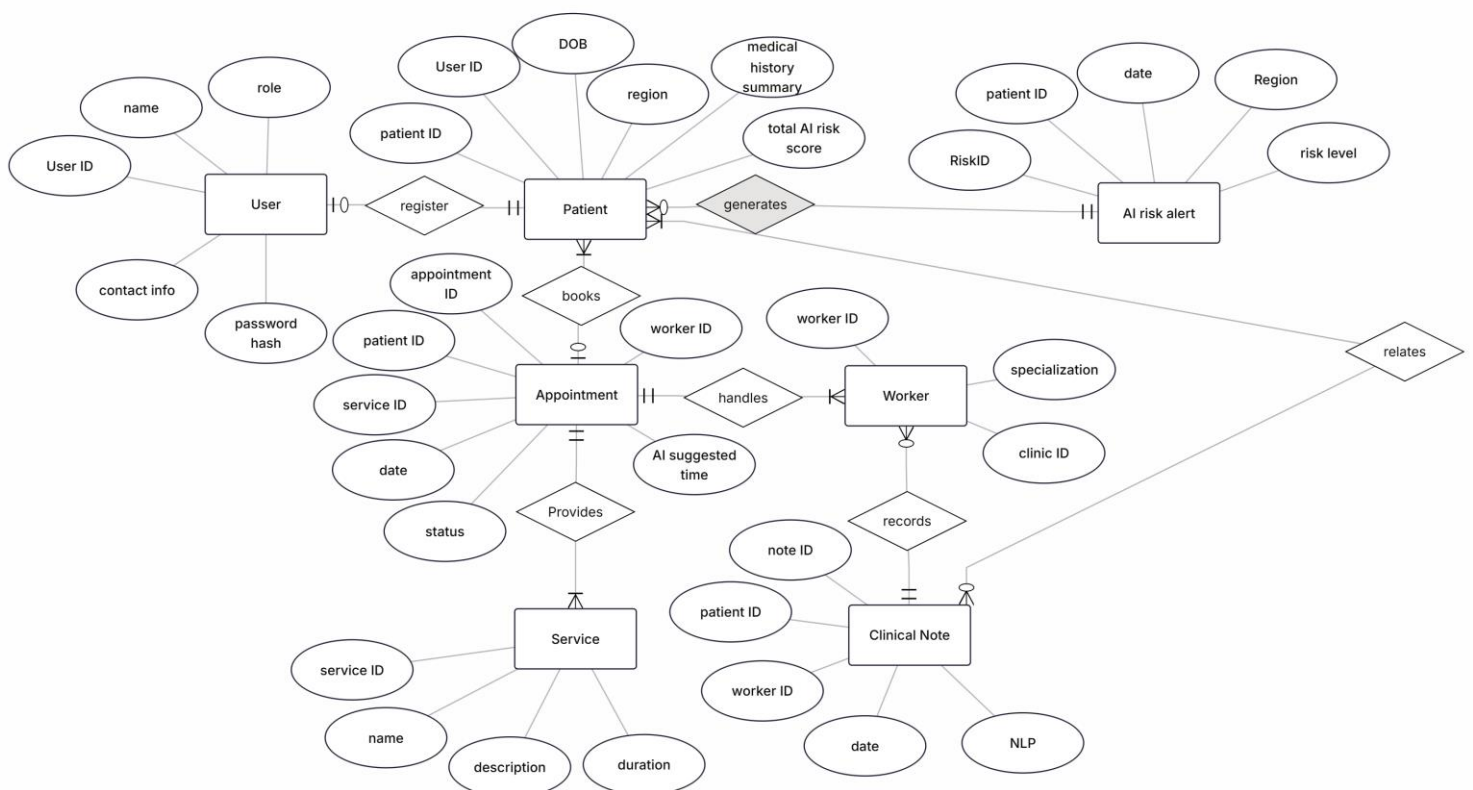


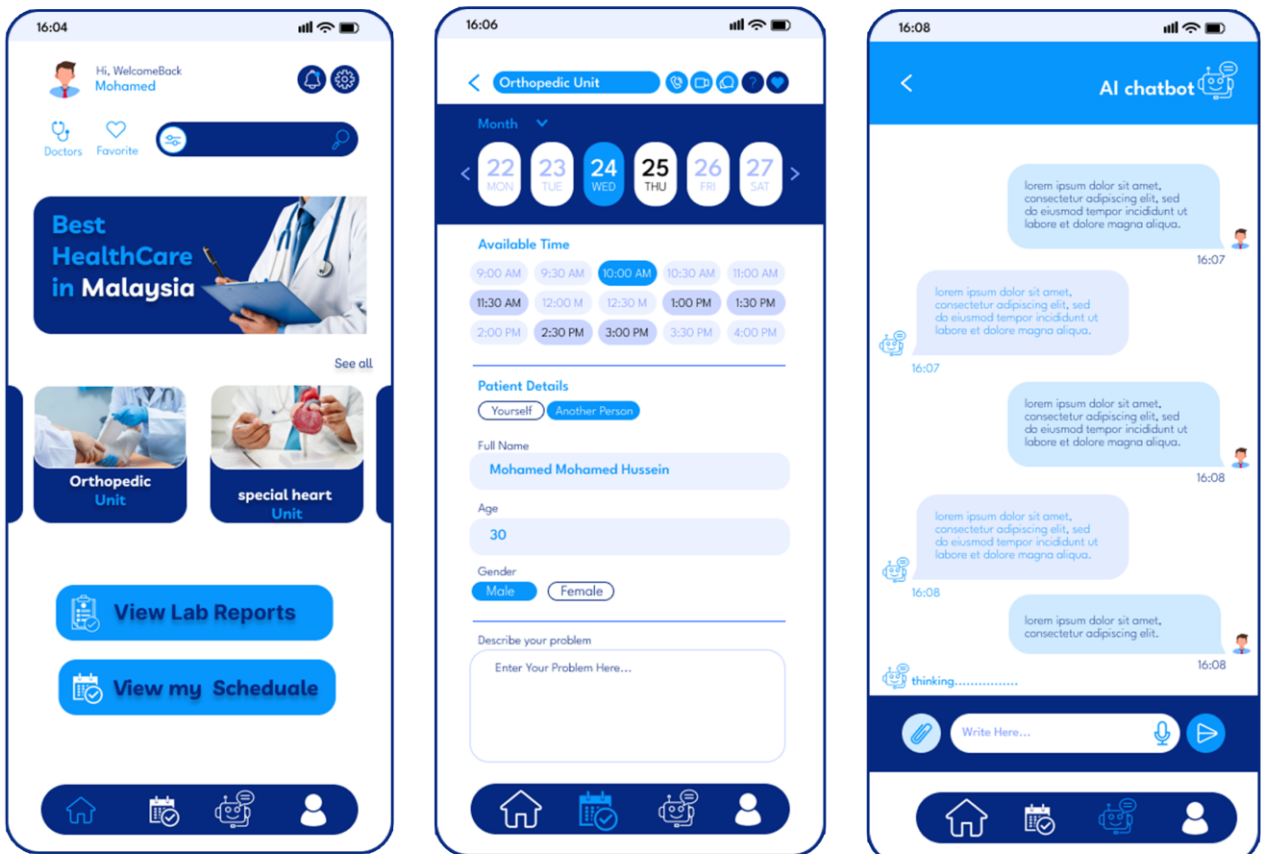
Table Name	Key Attributes (Focus)	AI-Specific Fields
PATIENT	PatientID (PK), DateOfBirth, Region	Total_AI_Risk_Score
APPOINTMENT	ApptID (PK), PatientID (FK), WorkerID (FK), Date	AI_Suggested_Time
CLINICAL_NOTE	NoteID (PK), PatientID (FK), WorkerID (FK), DateCreated	NLP_Transcription_Text, NLP_Confidence_Score
AI_RISK_ALERT	RiskID (PK), PatientID (FK), DateCalculated	Risk_Level, Model_Version_Used

5. Sample UI Wireframes

We designed 6 mockups to show the interface for all three user types, emphasizing the AI integration.

5.1 Citizen Module (Mobile View)

- **Screens:** Home Dashboard and Schedule Screen.
- **AI Feature:** The Schedule screen highlights one slot as the **AI's Top Recommendation**, making booking smart and easy. The Home screen has a clear **Chatbot icon** for instant information.



5.2 Healthcare Worker Module (Desktop/Tablet View)

- **Screens:** Patient Record Table and New Consultation Notes Screen.
- **AI Feature:** The New Consultation Notes Screen has a "Start Voice Note" button. After dictation, it shows "Transcription Complete. AI Confidence Score: 95%" to confirm quality.

HEALTHCARE WORKER

Dashboard

Patient Record

NLP Control

AI Feedback

Patient Records

Patient Record

Over View

Personal Details

Peterson James Davbhs

Home - 77 Milltown Rd A1, East Brunswick, New Jersey - 08816, United States.

james.peter@gmail.com

634 345 2366 | Mobile

DL58323873 - Drivers license

PAS5993234 - Passport

AB +

English, French

SSN 987675653423

Appointment

28 AUG

WED 11:00AM - 11:30 AM

Up Coming

with Dr. Raymond Gibbons

Contacts

Person name

Relation : Father

M 634 345 2366

fatheremail@gmail.com

Care giver name

Relation : Grandmother or Grandfather

M 634 345 2366, T 634 345 2366 | Work | 1221

caregivermail@gmail.com

Guarantor Name

Relation : Father

M 634 345 2366, T 634 345 2366 | Work | 1221

guarantormail@gmail.com

Payment Method

Insurance

Primary

Peterson James Davbhs

Generic Health Insurance

(843) 848-3483 | 0011 | Office

helpdesk@generichealthinsurance.com

Insurance

Secondary

Peterson James Davbhs

Generic Health Insurance

(843) 848-3483 | 0011 | Office

helpdesk@generichealthinsurance.com

Provider

Primary

Printo Thomas

M 868 345 2341

PrintoThomas@gmail.com

Home - 77 Milltown Rd Ste A1, East Brunswick, New Jersey - 08816, United States.

Dr. Secondary

Secondary

Dr. Secondary

M 868 345 2341

Secondaryprovider@gmail.com

Home - 77 Milltown Rd Ste A1, East Brunswick, New Jersey - 08816, United States.

Visit History

Date	Visit No.	Visit Status	Claim No.	Claim Status	Invoice No.	Inv. Amount	Inv. Balance	Sales Rep. Name
03/15/2019 10:20 AM	478628	Regular checkup	67654	Active	67823	\$ 1789	\$ 789	Murali Krishna B
03/15/2019 10:20 AM	New	Fever	56534	Closed	87654	\$ 2431	0	Raja Madem

Print Prescription

Save Record

Dictate Note

HEALTHCARE WORKER

Dashboard

Patient Record

NLP Control

AI Feedback

New Clinical Consultation Notes

Patient Record

Print Prescription

Save Record

Dictate Note

Transcription Complete. AI Confidence Score: 95%

The patient, Jenna Hassan, presents with a persistent chief complaint and fatigue for the fatigue. Her temperature or the the oxgus ain elevated at 100.2°F, 100.2°F, 98%, Her oxygen saturation, 98%. She denies any chest pain, she not shortness of shoreath, the sbreadth, or sore thraf, or sore thorat. Auscultation reveals clear lung sounds bilaterally. A rapid flu test negative. rapid flu test negative.

Recommended rest, hydration, and over-ower ouf ovler teror cough supressants. Follow-up ant ahetuled in shere days.

Visit History

Visit History

Eashbures

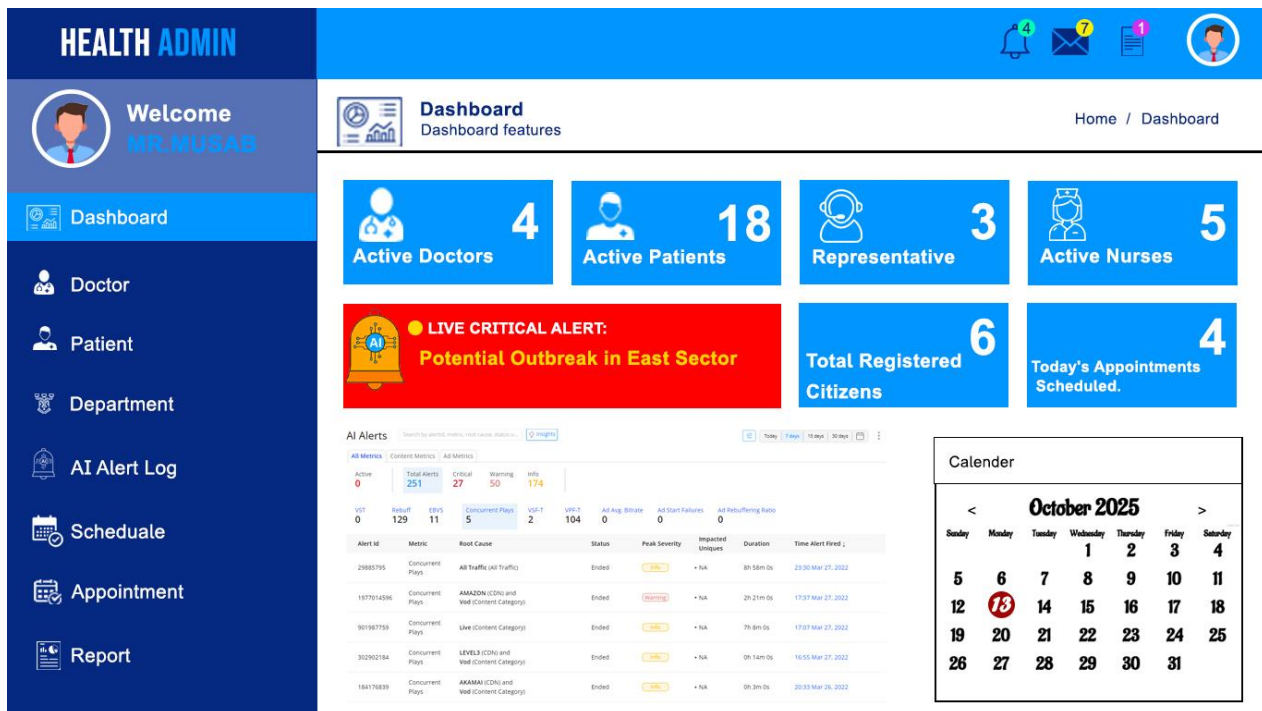
Consultation

Parlamde ant anpie the roit in Conte thes

View

5.3 Admin Staff Module (Desktop View)

- Screen: Main Dashboard.
- AI Feature: The dashboard features a large, unmissable red banner labeled "LIVE CRITICAL ALERT" showing the output of the Predictive Analytics Model. The sidebar includes a link to the AI Alert Log for history tracking.



6. Input Validation and Data Integrity Rules

We have these rules to ensure data is correct and safe, especially for the AI.

- **Required Fields:** All main fields (Name, Date of Birth) cannot be left empty.
- **Date Check:** Appointment dates must be in the future.
- **Role-Based Check:** Only a **Worker** can add a Prescription; the system checks their **Role ID**.
- **AI Confidence Check:** If the **NLP** model gives a low **Confidence_Score** (below 75%), the system flags the transcribed note, forcing the worker to review and correct it manually.
- **Anonymization:** Personal identifying data is removed before being sent to the **Predictive Analytics** service.

7. Deployment Strategy & Technology Stack Proposal

Component	Selection	Rationale (Why We Chose It)
Platform	Hybrid Mobile/Web	Web for Workers/Admin (large forms/dashboards); Mobile for Citizen (booking/chatbot).
Backend	Python (Django/Flask)	Python is the best language for AI/Machine Learning, making it easy to build and integrate our models.
Database	PostgreSQL	It is strong, secure, and excellent at maintaining the Data Integrity needed for sensitive health records (our ERD).
Hosting & AI Services	Google Cloud Platform (GCP)	GCP provides excellent scalability and offers the APIs we need, such as Google Cloud Speech-to-Text (for NLP) and Dialogflow (for the Chatbot).

8. AI System Strategy & Ethical Design

This section summarizes the AI operational plan and addresses all deployment, database, and ethical requirements.

8.1 AI Component Integration and Data Flow

This explains the full cycle of data for the AI features, showing where the system is driven by the intelligence.

- **NLP Flow (Data Flow):** The Record Controller sends the raw audio stream to the external GCP Speech-to-Text API. The API returns the text and the **Confidence Score**. The system then saves both the **transcribed text** and the **Confidence Score** to the **CLINICAL_NOTE** table, ensuring data integrity.
- **Predictive Analytics Flow (Data Flow):** The **Risk Analyzer Service** runs nightly, **pulls anonymized data** from the database, runs the AI model, and saves the result (the Risk Level) to the dedicated **AI_RISK_ALERT** table. This proves the system manages patient data separately from AI insights.
- **UI Integration (UI/Database):** The UI mockups visually confirm AI integration: the **Citizen App** shows AI-suggested booking times, and the **Admin Dashboard** displays the **LIVE CRITICAL ALERT** panel.

8.2 Deployment and Ethical Considerations

This confirms the viability and safety of the chosen architecture.

- **Deployment Plan:** Our strategy is built on **Google Cloud Platform (GCP)** to host scalable services like **GCP Speech-to-Text (for the NLP) and Dialog flow (for the Chatbot)**, which fulfills the requirement to mention cloud AI services.
- **Transparency & Consent:** The system adheres to ethical rules by ensuring:
 - **Transparency:** The Admin Dashboard shows the reason for the Live Critical Alert (e.g., "Predicted Flu Outbreak because of X symptom spike").
 - **Consent:** Citizens must give explicit permission for their anonymized data to be used for Predictive Analytics.
 - **Security:** All patient data is protected using encryption both when stored and when transmitted.